

Esercizio 1. (Alessandro Martina) Let $\varphi(x, y) \in L$, where $|x| = |y| = 1$. Suppose there is an infinite set $A \subseteq \mathcal{U}$ such that $\varphi(a, b) \leftrightarrow \varphi(b, a)$ for every two distinct $a, b \in A$. Prove that $\varphi(x; y)$ is unstable.

Esercizio 2. (Francesco Sulpizi) Find some unstable $p(x; z) \subseteq L(A)$ that admits no ladder of infinite length. Hint: work in a dense linear order, and consider the set of formulas of the form $x < y \vee y \notin \{a_0, \dots, a_n\}$.

Esercizio 3. (Leonardo Centazzo) Prove that

$$\bigcup \{\mathcal{M} : \mathcal{M} \text{ a minimal left ideal}\} = \bigcup \{\mathcal{M} : \mathcal{M} \text{ a minimal right ideal}\}.$$

Esercizio 4. (Roberto Carnevale) For every finite coloring of $\mathbb{N}$ and every $k < \omega$, there is an infinite sequence $\bar{a} = \langle a_i(x) : i < \omega \rangle$ of nonconstant univariate polynomials with coefficients in $\mathbb{N}$ such that $\{a(h) : a(x) \in \text{fp}(\bar{a}), h < k\}$ is monochromatic. Can we require that the $a_i(x)$ have increasing degrees?

Esercizio 5. (Matteo Bisi)

Esercizio 6. (Davide Peccioli) The following claim is too strong to be true. For every finite coloring of $\mathbb{N}^{(2)}$ there is an infinite sequence $\bar{a} = \langle a_i : i < \omega \rangle$ in $\mathbb{N}$ such that $\text{fp}(\bar{a})^{(2)}$ is monochromatic. Replace $\text{fp}(\bar{a})^{(2)}$ with a (slightly) smaller set that makes the claim true.

Esercizio 7. (Pietro Giura) Let $\mathcal{M}$ and $\mathcal{N}$ be type-definable minimal left ideals of $\mathcal{C}$. Let $u \in \mathcal{M}$ and $v \in \mathcal{N}$ be idempotents. Prove that $u * \mathcal{M}$ and $v * \mathcal{N}$ are isomorphic groups.

Esercizio 8. (Costanza Furone) Let $\mathcal{M}$ be a type-definable minimal left ideal of $\mathcal{C}$. Let $u \in \mathcal{M}$ be an idempotent. Prove that $u * \mathcal{M}$ is a minimal right ideal of $\mathcal{C}$.

Esercizio 9. (Angela Dosio) For every finite coloring of $\mathbb{N}$ and every $k < \omega$, there is an infinite sequence $\bar{a} = \langle a_i(x) : i < \omega \rangle$ of nonconstant univariate polynomials with coefficients in $\mathbb{N}$ such that $\{a(h) : a(x) \in \text{fp}(\bar{a}), h < k\}$ is monochromatic. Can we require that the $a_i(x)$ have increasing degrees?